Assignment 3

① Negative:

$L = 2^B = 2^4 = 16$

$S = L - 1 - r$

R	Eq	S
6	16-1-6	9
10	16-1-10	5
15	16-1-15	0
1	16-1-1	14
2	16-1-2	13
8	16-1-8	7
5	16-1-5	10
10	16-1-10	5
4	16-1-4	11
7	16-1-7	8
9	16-1-9	6
8	16-1-8	7
1	16-1-1	14
3	16-1-3	12
5	16-1-5	10
7	16-1-7	8

$$\begin{bmatrix} 6 & 10 & 15 & 1 \\ 2 & 8 & 5 & 10 \\ 4 & 7 & 9 & 8 \\ 1 & 3 & 5 & 7 \end{bmatrix}$$

original image

$$\begin{bmatrix} 9 & 5 & 0 & 14 \\ 13 & 7 & 10 & 5 \\ 11 & 8 & 6 & 7 \\ 14 & 12 & 10 & 8 \end{bmatrix}$$

Negative image

ⓑ Log:

$s = c \log_{10}(1+r)$

For simplicity: $c=1$

$$\begin{bmatrix} 6 & 10 & 15 & 1 \\ 2 & 8 & 5 & 10 \\ 4 & 7 & 9 & 8 \\ 1 & 3 & 5 & 7 \end{bmatrix}$$

original image

R	Eq	S
6	$1*\log_{10}(1+6)$	0.845
10	$1*\log_{10}(1+10)$	1.041
15	$1*\log_{10}(1+15)$	1.204
1	$1*\log_{10}(1+1)$	0.301
2	$1*\log_{10}(1+2)$	0.477
8	$1*\log_{10}(1+8)$	0.954
5	$1*\log_{10}(1+5)$	0.778
10	$1*\log_{10}(1+10)$	1.041
4	$1*\log_{10}(1+4)$	0.698
7	$1*\log_{10}(1+7)$	0.903
9	$1*\log_{10}(1+9)$	1
8	$1*\log_{10}(1+8)$	0.954
1	$1*\log_{10}(1+1)$	0.301
3	$1*\log_{10}(1+3)$	0.602
5	$1*\log_{10}(1+5)$	0.778
7	$1*\log_{10}(1+7)$	0.903

$\Longrightarrow$

R	Eq	S
6	$1*\log_{10}(1+6)$	1
10	$1*\log_{10}(1+10)$	1
15	$1*\log_{10}(1+15)$	1
1	$1*\log_{10}(1+1)$	0
2	$1*\log_{10}(1+2)$	0
8	$1*\log_{10}(1+8)$	1
5	$1*\log_{10}(1+5)$	1
10	$1*\log_{10}(1+10)$	1
4	$1*\log_{10}(1+4)$	1
7	$1*\log_{10}(1+7)$	1
9	$1*\log_{10}(1+9)$	1
8	$1*\log_{10}(1+8)$	1
1	$1*\log_{10}(1+1)$	0
3	$1*\log_{10}(1+3)$	1
5	$1*\log(1+5)$	1
7	$1*\log(1+7)$	1

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

Log Image

⊖ Power-Law (Gamma)

$S = cr^{\gamma}$

$c = 1$

$\gamma = 2$

$$\begin{bmatrix} 6 & 10 & 15 & 1 \\ 2 & 8 & 5 & 10 \\ 4 & 7 & 9 & 8 \\ 1 & 3 & 5 & 7 \end{bmatrix}$$

original image

R	Eq	S
6	$1*(6)^2$	36
10	$1*(10)^2$	100
15	$1*(15)^2$	225
1	$1*(1)^2$	1
2	$1*(2)^2$	4
8	$1*(8)^2$	64
5	$1*(5)^2$	25
10	$1*(10)^2$	100
4	$1*(4)^2$	16
7	$1*(7)^2$	49
9	$1*(9)^2$	81
8	$1*(8)^2$	64
1	$1*(1)^2$	1
3	$1*(3)^2$	9
5	$1*(5)^2$	25
7	$1*(7)^2$	49

R	Eq	S
6	$1*(6)^2$	15
10	$1*(10)^2$	15
15	$1*(15)^2$	15
1	$1*(1)^2$	1
2	$1*(2)^2$	4
8	$1*(8)^2$	15
5	$1*(5)^2$	15
10	$1*(10)^2$	15
4	$1*(4)^2$	15
7	$1*(7)^2$	15
9	$1*(9)^2$	15
8	$1*(8)^2$	15
1	$1*(1)^2$	1
3	$1*(3)^2$	9
5	$1*(5)^2$	15
7	$1*(7)^2$	15

$$\begin{bmatrix} 36 & 100 & 225 & 1 \\ 4 & 64 & 25 & 100 \\ 16 & 49 & 81 & 64 \\ 1 & 9 & 25 & 49 \end{bmatrix}$$

Power-Law Image

Convert to

$$\begin{bmatrix} 15 & 15 & 15 & 1 \\ 4 & 15 & 15 & 15 \\ 15 & 15 & 15 & 15 \\ 1 & 9 & 15 & 15 \end{bmatrix}$$

Power-Law Image

Because $L-1 = 15$ so we convert all new pixels to 15

(d) Thresholding:

$$S = \begin{cases} c & \text{if } r \geq m \\ 0 & \text{if } r < m \end{cases}$$

$c = 1$

$m = 7$

R	Eq	S
6	$6 \geq 7 \rightarrow$ False	0
10	$10 \geq 7 \rightarrow$ True	1
15	$15 \geq 7 \rightarrow$ True	1
1	$1 \geq 7 \rightarrow$ False	0
2	$2 \geq 7 \rightarrow$ False	0
8	$8 \geq 7 \rightarrow$ True	1
5	$5 \geq 7 \rightarrow$ False	0
10	$10 \geq 7 \rightarrow$ True	1
4	$4 \geq 7 \rightarrow$ False	0
7	$7 \geq 7 \rightarrow$ True	1
9	$9 \geq 7 \rightarrow$ True	1
8	$8 \geq 7 \rightarrow$ True	1
1	$1 \geq 7 \rightarrow$ False	0
3	$3 \geq 7 \rightarrow$ False	0
5	$5 \geq 7 \rightarrow$ False	0
7	$7 \geq 7 \rightarrow$ True	1

$$\begin{bmatrix} 6 & 10 & 15 & 1 \\ 2 & 8 & 5 & 10 \\ 4 & 7 & 9 & 8 \\ 1 & 3 & 5 & 7 \end{bmatrix}$$

original image

$$\begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Threshold Image

ⓒ Fixed Intensity Transformation:

$$S = \text{round}\left(\frac{1}{L-1} * r^2\right)$$

R	Eq	S
6	$\text{round}\left(\frac{1}{15} * 6^2\right)$	2
10	$\text{round}\left(\frac{1}{15} * 10^2\right)$	7
15	$\text{round}\left(\frac{1}{15} * 15^2\right)$	15
1	$\text{round}\left(\frac{1}{15} * 1^2\right)$	0
2	$\text{round}\left(\frac{1}{15} * 2^2\right)$	0
8	$\text{round}\left(\frac{1}{15} * 8^2\right)$	4
5	$\text{round}\left(\frac{1}{15} * 5^2\right)$	2
10	$\text{round}\left(\frac{1}{15} * 10^2\right)$	7
4	$\text{round}\left(\frac{1}{15} * 4^2\right)$	1
7	$\text{round}\left(\frac{1}{15} * 7^2\right)$	3
9	$\text{round}\left(\frac{1}{15} * 9^2\right)$	5
8	$\text{round}\left(\frac{1}{15} * 8^2\right)$	4
1	$\text{round}\left(\frac{1}{15} * 1^2\right)$	0
3	$\text{round}\left(\frac{1}{15} * 3^2\right)$	0
5	$\text{round}\left(\frac{1}{15} * 5^2\right)$	2
7	$\text{round}\left(\frac{1}{15} * 7^2\right)$	3

$$\begin{bmatrix} 6 & 10 & 15 & 1 \\ 2 & 8 & 5 & 10 \\ 4 & 7 & 9 & 8 \\ 1 & 3 & 5 & 7 \end{bmatrix}$$

original image

$$\begin{bmatrix} 2 & 7 & 15 & 0 \\ 0 & 4 & 2 & 7 \\ 1 & 3 & 5 & 4 \\ 0 & 0 & 2 & 3 \end{bmatrix}$$

Fixed intensity image

(F) Full Scale Contrast Stretch:

$r_{min} = 1$, $r_{max} = 15$, $L-1 = 15$

$$s = \text{round}\left((L-1) \cdot \frac{r - r_{min}}{r_{max} - r_{min}}\right)$$

$$\begin{bmatrix} 6 & 10 & 15 & 1 \\ 2 & 8 & 5 & 10 \\ 4 & 7 & 9 & 8 \\ 1 & 3 & 5 & 7 \end{bmatrix}$$

original image

R	Eq	S
6	$\text{round}\left(15 * \frac{6-1}{15-1}\right)$	5
10	$\text{round}\left(15 * \frac{10-1}{15-1}\right)$	10
15	$\text{round}\left(15 * \frac{15-1}{15-1}\right)$	15
1	$\text{round}\left(15 * \frac{1-1}{15-1}\right)$	0
2	$\text{round}\left(15 * \frac{2-1}{15-1}\right)$	1
8	$\text{round}\left(15 * \frac{8-1}{15-1}\right)$	8
5	$\text{round}\left(15 * \frac{5-1}{15-1}\right)$	4
10	$\text{round}\left(15 * \frac{10-1}{15-1}\right)$	10
4	$\text{round}\left(15 * \frac{4-1}{15-1}\right)$	3
7	$\text{round}\left(15 * \frac{7-1}{15-1}\right)$	6
9	$\text{round}\left(15 * \frac{9-1}{15-1}\right)$	9
8	$\text{round}\left(15 * \frac{8-1}{15-1}\right)$	8
1	$\text{round}\left(15 * \frac{1-1}{15-1}\right)$	0
3	$\text{round}\left(15 * \frac{3-1}{15-1}\right)$	2
5	$\text{round}\left(15 * \frac{5-1}{15-1}\right)$	4
7	$\text{round}\left(15 * \frac{7-1}{15-1}\right)$	6

$$\begin{bmatrix} 5 & 10 & 15 & 0 \\ 1 & 8 & 4 & 10 \\ 3 & 6 & 9 & 8 \\ 0 & 2 & 4 & 6 \end{bmatrix}$$

Full scale image